



Riemannian and Complex Geometry

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— Exercises —

4.1 Riemannian curvature

Exercise 4.1. Let M be a smooth manifold and E be a vector bundle on M . Let ∇ be a connection on E with curvature R defined as follows:

$$R: C^\infty(M, TM) \times C^\infty(M, TM) \times C^\infty(M, TM) \rightarrow C^\infty(M, TM)$$

$$(X, Y, s) \mapsto R(X, Y)s := \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X, Y]} s.$$

For any $X, Y \in C^\infty(M, TM)$, $s \in C^\infty(M, E)$ and $f_1, f_2, f_3 \in C^\infty(M)$. Prove that

$$R(f_1 X, f_2 Y)(f_3 s) = f_1 f_2 f_3 R(X, Y)(s).$$

In particular, the curvature R is a section of $T^*M \otimes T^*M \otimes \text{End } E$.

Exercise 4.2. Compute the Riemannian curvature, Ricci curvature and the scalar curvature of n -sphere, where the Riemannian metric on n -sphere is given in Exercise 3.4.

Exercise 4.3. Compute the Riemannian curvature, Ricci curvature and the scalar curvature of hyperbolic plane, where the Riemannian metric on hyperbolic plane is given in Exercise 3.5.

Exercise 4.4. Let (M, g) be a Riemannian manifold and T be a (r, s) -tensor, locally written as $T_{i_1 \dots i_r}^{j_1 \dots j_s} dx^{i_1} \otimes \dots \otimes dx^{i_r} \otimes \frac{\partial}{\partial x^{j_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{j_s}}$. Prove the Ricci identity

$$\nabla_{k,i}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} - \nabla_{i,k}^2 T_{i_1 \dots i_r}^{j_1 \dots j_s} = \sum_{\ell=1}^s R_{k i q}^{j \ell} T_{i_1 \dots i_r}^{j_1 \dots j_{\ell-1} q j_{\ell+1} \dots j_s} - \sum_{m=1}^r R_{k i i_m}^q T_{i_1 \dots i_{m-1} q i_{m+1} \dots i_r}^{j_1 \dots j_s}.$$

4.2 Taylor expansion of Riemannian metric

Exercise 4.5. Prove the following identities in normal coordinate:

1. $g^{ij} = \delta_{ij} + \frac{1}{3} R_{ik\ell j}(0) x^k x^\ell + O(|x|^3)$.
2. $\det(g_{ij}) = 1 - \frac{1}{3} R_{k\ell} x^k x^\ell + O(|x|^3)$.
3. $\sqrt{\det(g_{ij})} = 1 - \frac{1}{6} R_{k\ell} x^k x^\ell + O(|x|^3)$.

4.3 Einstein's Field Equations with Cosmological Constant

Exercise 4.6. Consider a 4-dimensional Lorentzian manifold $(\mathcal{M}, g_{\mu\nu})$ whose geometry satisfies the *Einstein vacuum field equations* with cosmological constant Λ :

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 0. \quad (4.1)$$

Here $R_{\mu\nu}$ is the Ricci tensor, $R = g^{\mu\nu}R_{\mu\nu}$ is the scalar curvature, and Λ is a real constant.

- (a) **(Einstein manifold condition)** Show that the field equations imply that $R_{\mu\nu} = 3k g_{\mu\nu}$ for some constant k , and determine k in terms of Λ .

Physical note: In vacuum, spacetime is an Einstein manifold. The constant k indicates whether the spacetime curvature is positive (de Sitter), negative (anti-de Sitter), or zero (flat).

- (b) **(Computation of the Christoffel symbols)** Consider a static, spherically symmetric spacetime with metric

$$ds^2 = -f(r) dt^2 + \frac{1}{f(r)} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (4.2)$$

Compute the Christoffel symbols $\Gamma_{\mu\nu}^\lambda$ in two independent ways:

- (i) Using their definition:

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2}g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}). \quad (4.3)$$

- (ii) Using the Euler–Lagrange equations for the Lagrangian

$$\mathcal{L} = \frac{1}{2}g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}, \quad (4.4)$$

where $g_{\mu\nu}$ is obtained from the metric above. Compare the equations of motion for each variable with the geodesic equation

$$\frac{d^2 x^\lambda}{d\tau^2} + \Gamma_{\mu\nu}^\lambda \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0, \quad (4.5)$$

and show that the Christoffel symbols obtained are consistent with those from part (i).

Physical note: This method shows that the motion of free particles follows geodesics determined by the Levi–Civita connection.

- (c) **(Computation of curvature tensors)** Compute the components of $R_{\mu\nu\lambda\rho}$, $R_{\mu\nu}$, and R for the metric in part (b) using two approaches:

$$R_{\sigma\mu\nu}^\rho = \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda,$$

- (i) From the coordinate expressions: $R_{\mu\nu} = g_{\rho\sigma} R_{\sigma\mu\nu}^\rho$,

$$R = g_{\mu\nu} R^{\mu\nu}.$$

- (ii) Using the differential form formalism: Express the connection 1-forms and curvature 2-forms, and follow the method outlined by David Tong in his notes (*Section 3.4.3: An Example: the Schwarzschild Metric*) to compute $R_{\mu\nu\lambda\rho}$, $R_{\mu\nu}$, and R .

Physical note: These tensors encode the intrinsic curvature of spacetime, describing how gravity manifests geometrically.

- (d) **(Determination of the metric function)** Assume that the metric in part (b) satisfies Einstein's field equations with cosmological constant Λ . Determine the function $f(r)$.

Physical note: The resulting metric is the Schwarzschild–(anti-)de Sitter solution, representing the gravitational field of a spherical mass in a universe with curvature determined by Λ .

- (e) **(Killing vectors and spacetime symmetries)** Determine all Killing vector fields for the spacetime obtained in part (d) for both cases $\Lambda > 0$ and $\Lambda < 0$.

Physical note: Killing vectors correspond to spacetime symmetries, such as time translation and rotational invariance, and lead to conserved quantities.

Summary:

Part	Mathematical Object	Physical Interpretation
(a)	Einstein manifold condition $R_{\mu\nu} = kg_{\mu\nu}$	Curvature set by Λ
(b)	Levi–Civita connection $\Gamma_{\mu\nu}^{\lambda}$	Geodesic motion of free particles
(c)	Riemann, Ricci, scalar curvature	Gravitational field geometry
(d)	Metric function $f(r)$	Schwarzschild–(A)dS geometry
(e)	Killing vectors	Symmetries and conserved quantities